

KARTHIK TATINENI

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EDUCATION

B.Tech in Electronics and Communication Engineering

2022 – 2026

Institute of Aeronautical Engineering, Hyderabad | CGPA: 7/10

SKILLS

Languages	Python, SQL
ML/MLOps	Algorithms(Regression, Classification, Clustering, Tree-Based/Ensemble), Feature Engineering, Evaluation Metrics, SHAP, Optuna, MLflow, Deep Learning (CNNs, RNN, LSTM, Transformers, YOLO), EDA, Statistics analysis
AI/GenAI	RAG, LLMs, Embeddings, Vector Databases, AI Agents, MCP, LangChain
Frameworks	FastAPI, Next.js
Databases	PostgreSQL, pgvector, MongoDB, MySQL, Redis, Firebase
Cloud	AWS (EC2, S3), Azure (Container Apps, AI Foundry), Docker, Cloudflare, Grafana
Tools	Pandas, NumPy, Matplotlib, Seaborn, PowerBI, Git, Postman, Hugging Face, Ollama, NVIDIA NIM, JupyterLab, Anaconda, pytest, Claude Code CLI

EXPERIENCE

Freelance Software Developer

Jan 2025 – Present

- Built **KLVORA Fashion Hub**, a production-ready Next.js e-commerce platform with Supabase, AI support, Three.js, Google SEO integration, secure authentication, inventory management, and order tracking. [[Live](#)]
- Built **Touchwood Interiors CMS**, a production-ready Next.js platform with Sanity CMS, Supabase, AI support chatbot, Google SEO integration, secure quotations, interactive 3D visualization, and architect portfolios. [[Live](#)]

PROJECTS

Free Trial Fraud Detection - Risk Scoring Engine | *Python, XGBoost, FastAPI, Redis, Optuna, MLflow, Docker* | [[GitHub](#)] [[Live Demo](#)]

- Engineered an end-to-end ML fraud detection pipeline utilizing causal feature engineering and an Incremental Union-Find entity graph; trained an Optuna-tuned XGBoost model calculating real-time risk scores (0–100) with **0.941 ROC-AUC**, **93.8% recall**, and **91.2% precision**.
- Deployed as a multi-tenant SaaS platform enabling developers to use API keys for real-time fraud scoring requests in <15ms, built with FastAPI, Redis, and Docker.

Aegis AI - Local-First AI Assistant Platform | *Python, PyTorch, Transformers, FastAPI, ChromaDB, RAG, LLMs* | [[GitHub](#)] [[Live Demo](#)]

- Designed and deployed a local-first Generative AI assistant using open-source LLMs (Mistral 7B, Qwen 2.5) with PyTorch and Hugging Face Transformers, optimizing GPU inference on an NVIDIA T4 instance via Lightning AI.
- Built a Retrieval-Augmented Generation (RAG) pipeline with a ChromaDB vector database and embedding-based semantic search, giving the assistant long-term memory for context-aware, multi-turn conversations.

Real-Time Helmet & Triple-Ride Detection | *Python, YOLO, CNN, TensorFlow, OpenCV, Tkinter* | [[GitHub](#)] [[Live Demo](#)]

- Developed a real-time computer vision pipeline using custom-trained YOLO and CNN deep learning models for multi-class object detection (helmet compliance, number-plate recognition) with TensorFlow and OpenCV.
- Deployed the trained model as a web application for live image-upload inference, automating end-to-end violation reporting with Excel logging and email alerts.

PUBLICATIONS

T. Karthik Sai et al., **Helmet and Triple-Ride Detection Using YOLO and CNN**, ICMCER 2026 (Scopus Indexed) | [Certificate](#)

CERTIFICATIONS

- Microsoft Certified(AI-901): Azure AI Fundamentals | [Certificate](#)
- Microsoft Certified(AI-103): Azure AI Apps and Agents Developer Associate | [Certificate](#)
- Microsoft Certified(DP-700): Fabric Data Engineer Associate | [Certificate](#)
- MySQL & Database Management: Create, Manage & Query Databases | [Certificate](#)
- HTML & CSS: The Complete Web Development Guide | [Certificate](#)

EXTRA-CURRICULAR

- Hosted and organised the official **IARE Consortium** event for 1,000+ participants; designed and developed the end-to-end registration platform.
- **Core Member and Logistics Head**, NSS - led community logistics for college and district level programs.